The Sen API

Interface Control Document

An example, generated from the data model by Sen. The reference that follows — every package, every type, every table — is written by the tool and is rewritten whenever the model changes.

This page is not. It is an ordinary Typst file included at a point the generated document leaves open, which is how a project puts its own cover, approvals and appendices around a reference it never has to edit.

Placeholder for your own project or product assets

Contents

About this document

This is an example published with Sen's documentation. It was produced by running the model through `sen generate typst` and compiling the result, with no editing in between.

Everything from *Model overview* onwards is generated. It follows the model and only the model: if a type is added, renamed or given a new description, the next build says so, and nothing anyone wrote by hand is lost in the process.

This page, and the cover before it, are not generated. They are Typst files of your own that the document includes at points it leaves open for you:

`--front-matter`	A cover, before the contents.
`--before-reference`	Anything between the contents and the model — a scope, a set of conventions, an approvals table. This page.
`--after-reference`	Appendices, after the model.
`--style`	Every decision about how the document looks.

The generator does not read any of those files. It writes an `#raw("#include")` and Typst resolves it when the document is compiled, so a cover page can be anything Typst can set — and changing it never means regenerating the model.

Model overview

335 types in 13 packages. 13 classes in 11 independent hierarchies.

Packages

Package	Types	Object classes	Fixed records	Enumerations	Variant records	Arrays	Optionals
components	—	—	—	—	—	—	—
ether	31	—	20	1	3	4	3
explorer	1	—	1	—	—	—	—
influx	3	—	1	1	—	1	—
jsonrpc	41	2	17	—	1	8	13
logmaster	3	2	1	—	—	—	—
py	3	1	1	—	—	1	—
recorder	6	1	2	1	—	2	—
replayer	4	2	1	1	—	—	—
rest	37	—	20	5	—	9	3
shell	27	1	20	3	1	2	—
db	3	—	3	—	—	—	—
kernel	160	4	80	23	8	37	8
log	16	—	11	1	1	3	—

Class hierarchy across all packages

Reading the tables

Every property carries three flags, in this order.

Flag	Meaning
`RO`	Read-only: the value is published by whoever owns it and cannot be set from outside.
`RW`	Read-write: the value can be set as well as read.
`D`	Dynamic: the value changes over the life of an instance.
`S`	Static: the value is fixed once an instance exists.
`C`	Confirmed: delivery is acknowledged, and the update is retried until it arrives.
`BE`	Best effort: the update is sent once and may be lost.

components

ether

31 types.

Fixed records

BusConfig STRUCTURE

Remember to set these to the same value in all related Sen instances

Name	Type	Description
`multicastRange`	MulticastRange	Defaults to the Organization Local scope (RFC 2365): [239], [192-195], [0-255], [0-255] Remember to keep a range wide enough for avoiding collisions if you have many buses.
`multicastExclusions`	MulticastAddress Exclusions	Address ranges that must not be allocated
`multicastPort`	u16	Defaults to 50985 Change this in case you have limited control over the infrastructure.
`multicastDisabled`	bool	If true, no multicast traffic is performed and all the bus settings above become meaningless. Defaults to false.

Named by Configuration.

BusJoined STRUCTURE

Name	Type	Description
`participantId`	u32	
`busId`	u32	
`busName`	string	

Named by ControlMessage.

BusLeft STRUCTURE

Name	Type	Description
`participantId`	u32	
`busId`	u32	

Name	Type	Description
busName	string	

Named by ControlMessage.

ByteRange STRUCTURE

Range of values of a u8

Name	Type	Description
min	u8	minimum range
max	u8	needs to be >= min

Named by MulticastRange.

Configuration STRUCTURE

Overall component configuration

Name	Type	Description
discovery	DiscoveryConfig	Discovery-related parameters
busOutQueue	QueueConfig	Queue for broadcasting to buses
processUdpOutQueue	QueueConfig	Queue for communicating with processes over UDP
processTcpOutQueue	QueueConfig	Queue for communicating with processes over TCP
busConfig	BusConfig	Bus transport configuration
portExclusions	PortExclusions	Port ranges that this process must not use
portConfig	PortConfig	If not present, all port bindings are Ephemeral
runDiscoveryHub	u16	If present, run the TCP discovery hub on this port
networkDevice	string	If present, the network device to use

DiscoveryHubAddress STRUCTURE

TCP-based discovery hub parameters

Name	Type	Description
port	u16	Port where the discovery hub is running
host	string	name or IP of the host where the discovery hub is running

Named by TcpDiscovery.

Endpoint STRUCTURE

Name	Type	Description
ip	u32	
port	u16	

Named by EnpointList.

Ephemeral STRUCTURE

Let the operating system select an ephemeral port

Named by PortBinding.

Hello STRUCTURE

Name	Type	Description
info	sen.kernel.ProcessInfo	

Name	Type	Description
udpPort	u16	
version	ProtocolVersion	

Named by ControlMessage.

MulticastAddressRange STRUCTURE

Inclusive IPv4 multicast address range

Name	Type	Description
min	string	
max	string	

Named by MulticastAddressExclusions.

MulticastDiscovery STRUCTURE

Multicast-based discovery parameters

Name	Type	Description
beamPeriod	sen.Duration	How often are we beaming our presence
beamExpirationTime	sen.Duration	Expiry time used by the beam tracker
mcastGroup	string	Defaults to '239.255.0.44'
port	u16	Defaults to 60543
device	string	Device to use for the network communication
allowVirtualInterfaces	bool	If true, virtual (NO_CARRIER) interfaces are also considered

Named by DiscoveryConfig.

PinnedPort STRUCTURE

Bind exactly this port, fail closed if taken

Name	Type	Description
port	u16	

Named by PortBinding.

PortConfig STRUCTURE

This configuration is per process and does not need to match across Sen instances. Exclusions apply to a pinned port, not to an ephemeral one: an ephemeral bind asks the OS for any free port and does not learn which it was given. udpUnicast cannot be pinned. Its socket is per peer and UDP has no four-tuple to demultiplex on, so a second peer either fails to bind or silently takes the first peer's traffic.

Name	Type	Description
tcpAcceptor	PortBinding	inbound TCP listener
udpUnicast	PortBinding	per-peer UDP unicast; must be Ephemeral
tcpSource	PortBinding	outbound TCP source port

Named by Configuration.

PortRange STRUCTURE

Inclusive port range

Name	Type	Description
min	u16	

Name	Type	Description
max	u16	

Named by PortExclusions.

ProbePortRange STRUCTURE

Bind within [min,max], random start, walk forward

Name	Type	Description
min	u16	
max	u16	

Named by PortBinding.

ProtocolVersion STRUCTURE

Name	Type	Description
kernel	u32	
ether	u32	

Named by Hello.

QueueConfig STRUCTURE

Name	Type	Description
evictionPolicy	QueueEvictionPolicy	What to do when a queue is full
maxSize	u64	Maximum element count, 0 means unbounded
warningLevel	u64	Element count where warning will be issued, 0 means none

Named by Configuration.

Ready STRUCTURE

Named by ControlMessage.

SessionPresenceBeam STRUCTURE

Name	Type	Description
protocolVersion	u16	
info	sen.kernel.ProcessInfo	
beamPeriod	sen.Duration	
endpoints	EnpointList	

TcpDiscovery STRUCTURE

TCP-based discovery parameters

Name	Type	Description
beamPeriod	sen.Duration	How often are we beaming our presence
beamExpirationTime	sen.Duration	Expiry time used by the beam tracker
hubAddress	DiscoveryHubAddress	Where to find the discovery hub
allowVirtual Interfaces	bool	If true, virtual (NO_CARRIER) interfaces are also considered

Named by DiscoveryConfig.

Enumerations

QueueEvictionPolicy ENUMERATION

What to do when a queue is full

Held as u32.

dropOldest	0
dropNewest	1

Named by QueueConfig.

Variant records

ControlMessage VARIANT

Type	Description
Hello	
Ready	
BusJoined	
BusLeft	

DiscoveryConfig VARIANT

Configuration parameters related to the discovery protocol

Type	Description
MulticastDiscovery	
TcpDiscovery	

Named by Configuration.

PortBinding VARIANT

Type	Description
Ephemeral	
PinnedPort	
ProbePortRange	

Named by PortConfig.

Arrays

EnpointList SEQUENCE

```
sequence<Endpoint, 27> EnpointList;
```

Named by SessionPresenceBeam.

MulticastAddressExclusions SEQUENCE

```
sequence<MulticastAddressRange> MulticastAddressExclusions;
```

Named by BusConfig.

MulticastRange SEQUENCE

Range of values of a IPv4 multicast address

```
array<ByteRange, 4> MulticastRange;
```

Named by BusConfig.

PortExclusions SEQUENCE

```
sequence<PortRange> PortExclusions;
```

Named by Configuration.

Optionals

MaybeDeviceName OPTIONAL

```
optional<string> MaybeDeviceName;
```

MaybeDiscoveryHubPort OPTIONAL

```
optional<u16> MaybeDiscoveryHubPort;
```

MaybePortConfig OPTIONAL

```
optional<PortConfig> MaybePortConfig;
```

explorer

1 types.

Fixed records

`Configuration` STRUCTURE

Overall component configuration

Name	Type	Description
`layoutFile`	string	the imgui layout file. Defaults to ""

influx

3 types.

Fixed records

Configuration STRUCTURE

Overall component configuration

Name	Type	Description
`samplingPeriod`	sen.Duration	sampling period for the recordings
`selections`	SelectionList	objects to be monitored (using sql)
`telegrafAddress`	string	ip or name of the Telegraf server
`telegrafPort`	u16	port used by the Telegraf server
`protocol`	Protocol	protocol used
`batchSize`	u32	points to batch before emission (0 means no batching)

Enumerations

Protocol ENUMERATION

Held as u8.

udp	0
tcp	1

Named by Configuration.

Arrays

SelectionList SEQUENCE

```
sequence<string> SelectionList;
```

Named by Configuration.

jsonrpc

41 types.

Object classes

JsonRpcServer CLASS

JsonRpcServer -- the JSON-RPC over WebSocket external API, expressed as a Sen class. One instance per connected client lives on `local.jsonrpc`. Methods are the RPC surface; events are the server-pushed notifications. Wire framing (envelope, id, error mapping) is the dispatcher's job, not this class's. Wire-shape convention: JSON-RPC named params (`"params": { ... }`) use the same field names as the STL method's argument list. The dispatcher treats every method's params object as `{ <argName>: <value>, ... }` and walks `method->getArgs()` to build the VarList in declaration order. That lets the inbound dispatch path go through Sen's meta machinery with no per-method translation.

Name	Type	Flags	Description
identity	Identity	RO S BE	Identity of the connected client.

Method	Signature	Description
ping	() → string	Liveness probe.
listTopology	() → SessionInfoList	Snapshot of the currently-detected sessions, each with the buses currently detected inside it. The kernel's discovery surface is asynchronous; this returns whatever was visible at the moment of the call. Use `subscribeTopology` for live updates.
subscribeTopology	()	Sticky subscription to topology changes. Idempotent per connection. Wires a `topologyChanged` push immediately with the current full snapshot, then again on every session or bus add/remove. Cancel with `unsubscribeTopology`; the subscription is also torn down automatically when the connection closes.
unsubscribeTopology	()	Cancel the topology subscription created by `subscribeTopology`. Idempotent.
getTypes	() → sen.kernel.StringList	Returns the qualified names of every custom type registered with the kernel. Catalog only -- pair with `getType` to fetch a chosen spec. The bundled-with-objects path on `interestUpdate.types` remains the primary way clients learn about types they observe.

Method	Signature	Description
getType	(qualifiedName: string, withSchema: bool) → TypeLookupResult	Look up one type by qualified name. Returns the CustomTypeSpec; when `withSchema` is true, also returns the JSON-Schema fragment for the type (empty string otherwise). Dependency chasing is the client's job: each schema fragment carries `$ref` strings keyed by qualified name, resolved against whatever schema cache the client maintains.
createInterest	(interestName: string, query: string, subscribe: SubscribeBlock, withSchemas: bool)	Register a sticky interest defined by an STL query. Optional `subscribe` block pre-subscribes to properties/events on every matched object, including ones that arrive later. When `withSchemas` is true, every subsequent `interestUpdate` for this interest carries the JSON-Schema fragment for each previously-unseen type alongside the existing `CustomTypeSpec`. Re-registering the same name = invalidParams. Wire-ordering contract: the initial `interestUpdate.added` for the current match set is emitted *before* this RPC's response lands on the wire. Clients must accept pushes for an interest whose response has not yet been observed.
releaseInterest	(interestName: string)	Drop an interest and every subscription living under it.
listObjects	(interestName: string) → ObjectInfos	List the names and classes of objects currently in an interest's match set.
getProperty	(interestName: string, objectName: string, propertyName: string) → string	Read a property's current value, synchronously.
getObjectsBatchState	(interestName: string, objectNames: sen.kernel.StringList, propertyNames: sen.kernel.StringList) → ObjectStateList	Batched per-object snapshot. Walks the interest's match set in one pass instead of forcing the caller to issue N*M `getProperty` round-trips. Omitting `objectNames` reads every object currently matched; omitting `propertyNames` reads every property of each object (the class hierarchy is walked, so inherited members are included). Names from `objectNames` that aren't currently in the match set are silently skipped -- the caller compares its requested list against the returned `objectName`s to detect omissions. Names from `propertyNames` that don't exist on a given object's class are recorded per-object under `errors`, so a heterogeneous batch still returns useful partial results. Bounded: the call rejects with `invalidParams` when the resolved object set exceeds 512 entries (either via an explicit

Method	Signature	Description
		`objectNames` list of that length, or via an implicit walk over a match set that has grown that large). Page the request by passing `objectNames`.
`setProperty`	`(interestName: string, objectName: string, propertyName: string, value: string)`	Write a property's value (sugar over invoking the generated setter). Hand-routed in `Dispatcher::handleSetProperty`: the dispatcher issues `invokeUntyped` against the property's setter and wires the response through the kernel callback in a single tick. The typed `setPropertyImpl` override in the C++ `Server` exists only to satisfy the codegen-generated abstract base and is unreachable at runtime. See `architecture.md` "Method dispatch" for why this method (and `invoke`) bypass the generic meta path.
`subscribeProperty`	`(interestName: string, objectName: string, propertyName: string, maxRateHz: f64)`	Subscribe to propertyChanged notifications for one (interest, object, property). Sticky: persists across object churn; auto-rewires when the named object (re)enters the match set. Snapshot of the current value is delivered immediately after wiring. The throttle is per-object, shared across every property/event subscribed for the same object. An explicit `maxRateHz` overrides the current rate for the object; omitting it preserves whatever rate is already in effect (so adding a per-property sub doesn't accidentally retune existing subs back to unlimited).
`unsubscribeProperty`	`(interestName: string, objectName: string, propertyName: string)`	Cancel a property subscription previously created with subscribeProperty.
`subscribeEvent`	`(interestName: string, objectName: string, eventName: string)`	Subscribe to eventTriggered notifications for one (interest, object, event).
`unsubscribeEvent`	`(interestName: string, objectName: string, eventName: string)`	Cancel an event subscription previously created with subscribeEvent.
`subscribeAll`	`(interestName: string, objectName: string, maxRateHz: f64)`	Subscribe to every property and every event of one named object. Wildcard expansion against the object's actual class (including parent classes). Object must currently be in the match set; absent objects are invalidParams.
`unsubscribeAll`	`(interestName: string, objectName: string)`	Cancel an all-members subscription previously created with subscribeAll.
`invoke`	`(interestName: string, objectName: string, methodName: string, argsJson: string) → string`	Call a method on a kernel object and await its return value. Hand-routed in `Dispatcher::handleInvoke`: validates the args, issues `invokeUntyped`, and wires the response through the kernel callback in a single tick. The typed

Method	Signature	Description
		`invokeImpl` override in the C++ `Server` exists only to satisfy the codegen-generated abstract base and is unreachable at runtime. See `architecture.md` "Method dispatch".

Event	Payload	Description
`propertyChanged`	`(interestName: string, objectName: string, values: PropertyValueList, timestamp: sen.TimeStamp)`	Bundled per-object snapshot of changed properties. Per-object bundling, not per-property: any property change on one object lands in a single envelope.
`eventTriggered`	`(interestName: string, objectName: string, eventName: string, args: string, timestamp: sen.TimeStamp)`	An event fired on one object in one interest. The event signature is decided by the source object's class, not by JSON-RPC.
`interestUpdate`	`(interestName: string, added: AddedObjectList, removed: sen.kernel.StringList, types: sen.kernel.CustomTypeSpecList, typeSchemas: string)`	Match-set delta for one interest. The dispatcher tracks knownTypes / knownSchemas per connection and only re-sends specs / schemas new to the client.
`topologyChanged`	`(sessions: SessionInfoList)`	Full snapshot of the topology after a session or bus add/remove. Pushed once on `subscribeTopology` (with the current state) and again on every change. Snapshot form rather than deltas: topology changes are rare, and the snapshot keeps client logic trivial and idempotent.
`notificationsDropped`	`(count: u64)`	Sent once when outbound backpressure clears on this connection, reporting how many unreliable notifications were dropped while the buffer was above the high watermark. Emitted reliably on purpose: the notice about dropped messages must not be droppable. A client that receives this has an incomplete picture and cannot tell which updates it missed -- only that it missed some. The correct response is to re-read what it displays (getObjectsBatchState over the affected interests) rather than to wait for the next change, which may never come for a property that has stopped moving.

`StaticFileServer` CLASS

Server-wide registry of in-memory static-file bundles, served alongside the WebSocket+JSON-RPC surface by the same uWS app. Internal-only: lives on the configurable control bus (defaults to `local.jsonrpc_control`), not on the data-plane bus that hosts authenticated per-connection JsonRpcServer instances. The component publishes exactly one instance.

Name	Type	Flags	Description
`registeredBundles`	RegisteredBundleInfoList	RO D C	Currently-registered bundles. Externally read-only; the component mutates this whenever a bundle is registered or unregistered. Subscribe to receive notifications on every change.

Method	Signature	Description
`registerStaticBundle`	`(bundle: StaticBundle) → u64`	Register a bundle. The bundle is copied into the registry; the caller may release the argument

Method	Signature	Description
		afterwards. Routes are wired up asynchronously on the uWS thread; the returned bundleId is stable and may be used immediately with `unregisterStaticBundle`.
`unregisterStaticBundle`	`(bundleId: u64)`	Stop serving the bundle. The slot in the registry is nulled; the underlying uWS route remains registered (uWS does not expose a clean route-removal API) but becomes inert and returns 404. No-op if `bundleId` is unknown.

Fixed records

`AddedObjectEntry` STRUCTURE

One object newly entering an interest's match set.

Name	Type	Description
`objectName`	string	name of the newly-matched object
`qualifiedClassName`	string	package-qualified class name of the object
`currentValues`	PropertyValueList	initial values of pre-subscribed properties (empty if none wired)

Named by AddedObjectList.

Configuration STRUCTURE

JSON-RPC over WebSocket component configuration. Only `address` and `port` are required; every other field falls back to a documented default.

Name	Type	Description
address	string	The IP address or hostname the server will bind to
port	u16	The port number the server will listen on
updateFreqHz	f64	Run-loop frequency in Hz; defaults to 60
connectionLimits	ConnectionLimits	Per-connection uWS buffer / timeout knobs
tls	TlsConfig	schema placeholder; setting this fails component start
controlBusName	string	Control-plane bus on `local`; defaults to "jsonrpc_control"

ConnectionLimits STRUCTURE

Per-connection buffer / timeout knobs consulted on the uWS thread for the lifetime of the server. Every field is optional; unset fields fall back to the defaults resolved inside ws_server.cpp at the use site.

Name	Type	Description
idleTimeoutSeconds	u16	uWS idle close timeout; defaults to 120
maxPayloadBytes	u32	Max inbound frame size; defaults to 16 KiB
maxBackpressureBytes	u32	uWS hard outbound buffer ceiling; defaults to 64 KiB
highBackpressureBytes	u32	Soft trigger for the BackpressureUpdate signal; defaults to 32 KiB

Named by Configuration.

Identity STRUCTURE

Identity of the connected client. Set once on connect by the dispatcher from the WebSocket upgrade authenticator. Read-only externally. Other components / admin tools can SELECT * FROM local.jsonrpc to enumerate currently-connected clients.

Name	Type	Description
subject	string	authenticated subject identifier (e.g. user / service principal)

Named by JsonRpcServer.

NamedSelection STRUCTURE

Named arm of MemberSelector: matches only the listed members. Wire shape: `{type: "NamedSelection", value: {memberNames: [...]}}`.

Name	Type	Description
memberNames	sen.kernel.StringList	member names to match

Named by MemberSelector.

ObjectInfo STRUCTURE

One entry of listObjects' result.

Name	Type	Description
objectName	string	object name within the interest's match set
qualifiedClassName	string	package-qualified class name of the object

Named by ObjectInfos.

ObjectState STRUCTURE

Per-object snapshot returned by `getObjectsBatchState`. `properties` carries successful reads (one entry per property name resolved against the object's class hierarchy); `errors` is empty when nothing failed for this object.

Name	Type	Description
objectName	string	object name within the interest's match set
qualifiedClassName	string	package-qualified class name at read time
properties	PropertyValueList	resolved property values; empty when caller's filter matched nothing
errors	PropertyErrorList	per-property failures; empty when none occurred

Named by ObjectStateList.

PropertyError STRUCTURE

Per-property read error. Used when the caller-supplied `propertyNames` includes a name that doesn't exist on a given object's class, or when the accessor itself throws. Carried in a separate list from successful reads so a property literally named `error` can never collide with the failure envelope.

Name	Type	Description
propertyName	string	name from the caller's `propertyNames` list that could not be read
error	string	human-readable reason (e.g. "unknown property")

Named by PropertyErrorList.

PropertyValuePair STRUCTURE

Per-property changed value. The value type is decided by the addressed kernel object, not by JSON-RPC, so STL can't statically describe it.

Name	Type	Description
propertyName	string	property member name
value	string	JSON-encoded SenValue

Named by PropertyValueList.

RegisteredBundleInfo STRUCTURE

Summary record returned by `listRegisteredBundles`. Excludes the file bytes; just enough to answer "what is currently being served, and where".

Name	Type	Description
bundleId	u64	opaque id assigned by the server, stable for the bundle's lifetime
urlPrefix	string	bundle's URL prefix
fileCount	u64	number of files in the bundle

Named by RegisteredBundleInfoList.

SessionInfo STRUCTURE

One discovered session and the buses currently detected inside it. Names are opaque strings from the kernel's session-discovery surface.

Name	Type	Description
name	string	session name
buses	sen.kernel.StringList	bus names currently detected in this session

Named by SessionInfoList.

StaticBundle STRUCTURE

One bundle of in-memory files served on a configurable URL prefix. The same bundle's files are served at `<urlPrefix>/<file.path>`. `indexFileName` is the SPA fallback served whenever a request under `urlPrefix` does not match any file (lets the client-side router handle paths the server does not know about).

Name	Type	Description
`urlPrefix`	string	URL prefix, e.g. "/explorer"; must start with "/"
`indexFileName`	string	SPA fallback, e.g. "index.html"; must match a `path` in `files`
`files`	StaticFileList	file set

Named by StaticFileServer.

StaticFile STRUCTURE

One file inside a StaticBundle. Bytes are owned by the bundle; `path` is relative to the bundle's `urlPrefix` and uses forward slashes regardless of the producer's OS. `contentType` is the MIME type the server will return verbatim; the bundle producer decides it (no extension-based inference happens server-side).

Name	Type	Description
`path`	string	path within the bundle, e.g. "assets/main.js"
`contentType`	string	MIME type, e.g. "text/html"
`contents`	sen.kernel.Buffer	raw file bytes

Named by StaticFileList.

SubscribeBlock STRUCTURE

Pre-subscription block bundled into createInterest. Wires the listed property / event subscriptions on every matched object -- including ones that arrive later. Wire shape mirrors the typed STL representation exactly: each `MaybeMemberSelector` is either omitted (no auto-subscribe), `null`, or one of the typed variant arms (`{type: "WildcardSelection", value: {}}` or `{type: "NamedSelection", value: {memberNames: [...]}}`); `maxRateHz` is an optional positive number. Strict alignment with the typed surface lets the dispatcher's uniform meta-dispatch path consume the wire directly via `fromJson`, without a per-method translation step.

Name	Type	Description
`properties`	MemberSelector	which properties to auto-subscribe to (or none)
`events`	MemberSelector	which events to auto-subscribe to (or none)
`maxRateHz`	f64	optional throttle cap shared across the wired subscriptions

Named by JsonRpcServer.

TlsConfig STRUCTURE

TLS materials. The schema is reserved so configs stay forward-compatible, but the component fails to start if `tls` is set; full TLS support is not yet wired through the uWebSockets SSL backend.

Name	Type	Description
`certPath`	string	PEM-encoded server certificate
`keyPath`	string	PEM-encoded server private key

Named by Configuration.

TypeLookupResult STRUCTURE

Bundled return of getType: always the CustomTypeSpec, plus the JSON-Schema fragment when the caller opted in. `schema` is empty when not requested.

Name	Type	Description
`spec`	sen.kernel.CustomTypeSpec	

Name	Type	Description
`schema`	string	JSON-encoded schema fragment; "" when withSchema=false

Named by JsonRpcServer.

`WildcardSelection` STRUCTURE

Wildcard arm of MemberSelector: matches every member of the relevant kind. Wire shape: `{type: "WildcardSelection", value: {}}`.

Named by MemberSelector.

Variant records

`MemberSelector` VARIANT

Selector for properties or events in a SubscribeBlock. Absent = no auto-subscribe; WildcardSelection = "*"; NamedSelection = explicit list.

Type	Description
WildcardSelection	
NamedSelection	

Named by SubscribeBlock.

Arrays

`AddedObjectList` SEQUENCE

```
sequence<AddedObjectEntry> AddedObjectList;
```

Named by JsonRpcServer.

`ObjectInfos` SEQUENCE

```
sequence<ObjectInfo> ObjectInfos;
```

Named by JsonRpcServer.

`ObjectStateList` SEQUENCE

```
sequence<ObjectState> ObjectStateList;
```

Named by JsonRpcServer.

`PropertyErrorList` SEQUENCE

```
sequence<PropertyError> PropertyErrorList;
```

Named by ObjectState.

PropertyValueList `SEQUENCE`

```
sequence<PropertyValuePair> PropertyValueList;
```

Named by AddedObjectEntry, JsonRpcServer, ObjectState.

RegisteredBundleInfoList `SEQUENCE`

```
sequence<RegisteredBundleInfo> RegisteredBundleInfoList;
```

Named by StaticFileServer.

SessionInfoList `SEQUENCE`

```
sequence<SessionInfo> SessionInfoList;
```

Named by JsonRpcServer.

StaticFileList `SEQUENCE`

```
sequence<StaticFile> StaticFileList;
```

Named by StaticBundle.

Optionals

MaybeConnectionLimits `OPTIONAL`

```
optional<ConnectionLimits> MaybeConnectionLimits;
```

MaybeControlBusName `OPTIONAL`

```
optional<string> MaybeControlBusName;
```

MaybeFlag `OPTIONAL`

```
optional<bool> MaybeFlag;
```

MaybeHighBackpressureBytes `OPTIONAL`

```
optional<u32> MaybeHighBackpressureBytes;
```

MaybeIdleTimeoutSeconds OPTIONAL

```
optional<u16> MaybeIdleTimeoutSeconds;
```

MaybeMaxBackpressureBytes OPTIONAL

```
optional<u32> MaybeMaxBackpressureBytes;
```

MaybeMaxPayloadBytes OPTIONAL

```
optional<u32> MaybeMaxPayloadBytes;
```

MaybeMemberSelector OPTIONAL

```
optional<MemberSelector> MaybeMemberSelector;
```

MaybeRateHz OPTIONAL

```
optional<f64> MaybeRateHz;
```

MaybeStringList OPTIONAL

Optional list of object names. Absent on the wire (or `null`) selects every object currently in the addressed interest's match set; present limits the read to the listed names.

```
optional<sen.kernel.StringList> MaybeStringList;
```

MaybeSubscribeBlock OPTIONAL

```
optional<SubscribeBlock> MaybeSubscribeBlock;
```

MaybeTlsConfig OPTIONAL

```
optional<TlsConfig> MaybeTlsConfig;
```

MaybeUpdateFreqHz OPTIONAL

```
optional<f64> MaybeUpdateFreqHz;
```

logmaster

3 types.

Object classes

`LogMaster` CLASS

allows controlling the loggers

Name	Type	Flags	Description
`targetBus`	string	RO S BE	

Method	Signature	Description
`muteAll`	`()`	mute all the loggers
`unmuteAll`	`()`	un-mutes all the muted loggers
`toggleMuteAll`	`()`	toggles the muting of all loggers
`setLevel`	`(level: sen.kernel.log.LogLevel)`	sets all the loggers to a given level

`Logger` CLASS

represents a logger

Name	Type	Flags	Description
`level`	sen.kernel.log.LogLevel	RO D BE	the current log level of this logger
`pattern`	string	RO D C	the pattern used by this logger
`muted`	bool	RO D BE	true if muted

Method	Signature	Description
`mute`	`()`	mute this logger
`unmute`	`()`	unmute this logger
`setLevel`	`(level: sen.kernel.log.LogLevel)`	unmute this logger
`setPattern`	`(pattern: string)`	sets the pattern for this logger

Fixed records

`Config` STRUCTURE

component configuration

Name	Type	Description
`targetBus`	string	defaults to "local.log"
`period`	sen.Duration	defaults to 500 milliseconds

py

3 types.

Object classes

PythonInterpreter CLASS

represents the embedded Python interpreter

Method	Signature	Description
`eval`	`(expr: string) → string`	evaluates a Python expression
`exec`	`(code: string)`	executes a Python code block
`execFile`	`(file: string)`	executes a Python file

Fixed records

Configuration STRUCTURE

Overall component configuration

Name	Type	Description
`freqHz`	f32	update frequency in Hertz
`module`	string	python module to use (name of a python script, without the .py)
`imports`	StringList	sen packages to import
`bus`	string	bus where to publish the interpreter object (none by default)

Arrays

StringList SEQUENCE

```
sequence<string> StringList;
```

Named by Configuration.

recorder

6 types.

Object classes

Recorder CLASS

represents an recording session

Name	Type	Flags	Description
settings	RecordingSettings	RO S BE	
state	RecorderState	RO D C	

Method	Signature	Description
start	()	starts (or resumes) recording
stop	()	stops recording
fetchStats	() → sen.db.OutStats	sample the output statistics

Fixed records

Configuration STRUCTURE

Overall component configuration

Name	Type	Description
samplingPeriod	sen.Duration	sampling period for the recordings
recordings	RecordingSettingsList	predefined recordings
bus	string	bus to publish the object (local.rec by default)

RecordingSettings STRUCTURE

Configuration of a predefined recording

Name	Type	Description
name	string	name of the recording (cannot be empty)
folder	string	storage folder
selections	SelectionList	which objects shall be recorded
keyframePeriod	sen.Duration	period between keyframes (0 means no keyframes)
indexKeyframes	bool	should we create indexes for keyframes
indexObjects	bool	should we create indexes for objects
autoStart	bool	should we start recording automatically

Named by Recorder, RecordingSettingsList.

Enumerations

RecorderState ENUMERATION

the state of a recording session

Held as u8.

stopped	0	error	2
recording	1		

Named by Recorder.

Arrays

RecordingSettingsList SEQUENCE

List of predefined recordings

```
sequence<RecordingSettings> RecordingSettingsList;
```

Named by Configuration.

SelectionList SEQUENCE

List of selection queries

```
sequence<string> SelectionList;
```

Named by RecordingSettings.

replayer

4 types.

Object classes

Replay CLASS

controls the replay of a given archive

Name	Type	Flags	Description
`archiveInfo`	sen.db.Summary	RO S BE	information about the archive being replayed
`archivePath`	string	RO S BE	file path to the archive
`status`	ReplayStatus	RO D C	playback status
`playbackTime`	sen.TimeStamp	RO D BE	playback time

Method	Signature	Description
`play`	`()`	starts the playback from the current position
`pause`	`()`	hold the playback position, if playing
`stop`	`()`	goes to the start of the playback, erasing all objects
`seek`	`(time: sen.TimeStamp)`	restores the playback to the keyframe that's closest to a given time
`advance`	`(time: sen.Duration)`	if paused or stopped, advances the playback a given time delta

Replayer CLASS

represents the replayer component and manages Replay objects.

Name	Type	Flags	Description
`autoPlay`	bool	RO S BE	if true, automatically calls play when opening a replay

Method	Signature	Description
`open`	`(name: string, path: string)`	opens an archive in a given path, and creates a named Replay object.
`close`	`(name: string)`	closes the named Replay object, if any.
`closeAll`	`()`	closes all previously opened replays

Fixed records

Configuration STRUCTURE

Overall component configuration

Name	Type	Description
`bus`	string	bus to publish the object (local.replay by default)
`samplingPeriod`	sen.Duration	max sampling period for flushing the replayed data
`autoOpen`	string	optional archive path to auto open on start-up (empty by default)
`autoPlay`	bool	if autoOpen is not empty, try to play it on start-up (false by default)

Enumerations

`ReplayStatus` ENUMERATION

the state in which a Replay can be

Held as u8.

stopped	0
paused	1
playing	2

Named by Replay.

rest

37 types.

Fixed records

`Argument` STRUCTURE

Name	Type	Description
`name`	string	
`description`	string	
`type`	string	

Named by Arguments.

`Bus` STRUCTURE

Name	Type	Description
`id`	string	
`status`	Status	
`objects`	ObjectsSummary	

`BusSummary` STRUCTURE

Name	Type	Description
`id`	string	
`status`	Status	

Named by Buses.

`ClientAuth` STRUCTURE

Name	Type	Description
`id`	string	

`Configuration` STRUCTURE

Name	Type	Description
`address`	string	The IP address or hostname the server will bind to
`port`	u16	The port number the server will listen on
`freqHz`	f32	Update frequency in Hertz

`Error` STRUCTURE

Name	Type	Description
`error`	string	

`Interest` STRUCTURE

Name	Type	Description
`query`	string	
`name`	string	

`InterestSummary` STRUCTURE

Name	Type	Description
`name`	string	
`session`	string	
`bus`	string	

Named by InterestsSummary.

`Link` STRUCTURE

Name	Type	Description
`rel`	RelType	
`href`	string	
`method`	HttpMethod	

Named by Links, ObjectSummary.

`Notification` STRUCTURE

Name	Type	Description
`type`	NotificationType	
`interest`	string	
`time`	sen.TimeStamp	
`data`	string	

`Object` STRUCTURE

Name	Type	Description
`objectId`	u32	
`name`	string	
`className`	string	
`localName`	string	

Name	Type	Description
description	string	
links	Links	

ObjectEvent STRUCTURE

Name	Type	Description
name	string	
description	string	

ObjectMethod STRUCTURE

Name	Type	Description
name	string	
description	string	
retType	string	
args	Arguments	

ObjectSummary STRUCTURE

Name	Type	Description
objectId	u32	
name	string	
className	string	
localName	string	
link	Link	

Named by ObjectsSummary.

Session STRUCTURE

Name	Type	Description
name	string	

SessionSummary STRUCTURE

Name	Type	Description
name	string	
buses	BusNames	

SubscriptionOptions STRUCTURE

Name	Type	Description
maxUpdateTime	sen.Duration	Max. update time

Subscriptions STRUCTURE

Name	Type	Description
properties	Properties	
events	Events	

Success `STRUCTURE`

Name	Type	Description
`msg`	string	

Version `STRUCTURE`

Name	Type	Description
`version`	string	

Enumerations

HttpMethod `ENUMERATION`

Held as u8.

httpGet	0
httpPost	1
httpPut	2
httpDelete	3
httpOption	4

Named by Link.

InvokeStatus `ENUMERATION`

Held as u8.

pending	0
finished	1
failed	2

NotificationType `ENUMERATION`

Held as u8.

evt	0
invoke	1
property	2
objectAdded	3
objectRemoved	4
count	5

Named by Notification.

RelType `ENUMERATION`

Held as u8.

method	0
setter	1
getter	2
evt	3
property	4
propertySubscribe	5
propertyUnsubscribe	6
def	7

Named by Link.

Status ENUMERATION

Held as u8.

open	0
close	1

Named by Bus, BusSummary.

Arrays

Arguments SEQUENCE

```
sequence<Argument> Arguments;
```

Named by ObjectMethod.

BusNames SEQUENCE

```
sequence<string> BusNames;
```

Named by SessionSummary.

Buses SEQUENCE

```
sequence<BusSummary> Buses;
```

Events SEQUENCE

```
sequence<string> Events;
```

Named by Subscriptions.

InterestsSummary SEQUENCE

```
sequence<InterestSummary> InterestsSummary;
```

Links SEQUENCE

```
sequence<Link> Links;
```

Named by Object.

ObjectsSummary SEQUENCE

```
sequence<ObjectSummary> ObjectsSummary;
```

Named by Bus.

Properties SEQUENCE

```
sequence<string> Properties;
```

Named by Subscriptions.

Sessions SEQUENCE

```
sequence<string> Sessions;
```

Optionals

ThreadPoolSize OPTIONAL

```
optional<u16> ThreadPoolSize;
```

UpdateFrequency OPTIONAL

```
optional<f32> UpdateFrequency;
```

UpdateTime OPTIONAL

```
optional<sen.Duration> UpdateTime;
```

shell

27 types.

Object classes

Shell CLASS

Provides a text-based interface to interact with the a sen kernel

Name	Type	Flags	Description
config	Configuration	RO S BE	service configuration

Method	Signature	Description
clear	()	clears the screen
help	()	basic usage help
history	()	lists previous commands
ls	()	lists all the existing objects and sources.
open	(source: string)	opens a source (bus or session) without any condition
query	(name: string, condition: string)	opens a named source using a condition expression
close	(source: string)	closes a previously-opened source (session, bus or query)
info	(term: string)	prints information about a type or instance
shutdown	()	requests the kernel to be shut-down
src	()	lists the current sources

Fixed records

CPrint STRUCTURE

Name	Type	Description
flags	u32	
color	u32	
text	string	

Named by TerminalCmd.

`ClearCurrentLine` STRUCTURE

Named by TerminalCmd.

`ClearRemainingCurrentLine` STRUCTURE

Named by TerminalCmd.

`ClearScreen` STRUCTURE

Named by TerminalCmd.

`Configuration` STRUCTURE

Configuration params for the shell

Name	Type	Description
`prompt`	string	defaults to 'sen> '
`timeStyle`	TimeStyle	how to display time
`bufferStyle`	BufferStyle	how to display buffers
`serverEnabled`	bool	enables remote shell access (false by default)
`serverPort`	u16	0 means default
`noLogo`	bool	true if you don't want a welcoming logo
`open`	SourcesList	list of sources to open automatically
`query`	QueryList	list of queries to create automatically
`logBus`	string	where to find the logmaster, if any (default: local.log)

Named by Shell.

`HideCursor` STRUCTURE

Named by TerminalCmd.

`MoveCursorAllLeft` STRUCTURE

Named by TerminalCmd.

`MoveCursorDown` STRUCTURE

Name	Type	Description
`cells`	u32	

Named by TerminalCmd.

`MoveCursorLeft` STRUCTURE

Name	Type	Description
`cells`	u32	

Named by TerminalCmd.

`MoveCursorRight` STRUCTURE

Name	Type	Description
`cells`	u32	

Named by TerminalCmd.

MoveCursorUp STRUCTURE

Name	Type	Description
cells	u32	

Named by TerminalCmd.

NewLine STRUCTURE

Named by TerminalCmd.

Print STRUCTURE

Name	Type	Description
text	string	

Named by TerminalCmd.

Query STRUCTURE

Holds information about a named query

Name	Type	Description
name	string	the name of the query (cannot contain spaces or dots)
selection	string	the query expression

Named by QueryList.

RestoreCursorPosition STRUCTURE

Named by TerminalCmd.

SaveCursorPosition STRUCTURE

Named by TerminalCmd.

SetBgColor STRUCTURE

Name	Type	Description
color	u32	

Named by TerminalCmd.

SetFgColor STRUCTURE

Name	Type	Description
color	u32	

Named by TerminalCmd.

SetWindowTitle STRUCTURE

Name	Type	Description
text	string	

Named by TerminalCmd.

ShowCursor STRUCTURE

Named by TerminalCmd.

Enumerations

BufferStyle ENUMERATION

How buffers should be printed

Held as u8.

size	0
hexdump	1

Named by Configuration.

ListStyle ENUMERATION

How lists should be printed

Held as u8.

treeStyle	0
plainStyle	1

TimeStyle ENUMERATION

How timestamps should be printed

Held as u8.

timestampUTC	0
timestampLocal	1

Named by Configuration.

Variant records

TerminalCmd VARIANT

Type	Description
Print	
CPrint	
NewLine	
MoveCursorLeft	
MoveCursorRight	
MoveCursorUp	
MoveCursorDown	
MoveCursorAllLeft	
SetFgColor	
SetBgColor	
ClearCurrentLine	
ClearRemainingCurrentLine	
HideCursor	
ShowCursor	
SaveCursorPosition	

Type	Description
RestoreCursorPosition	
ClearScreen	
SetWindowTitle	

Arrays

QueryList SEQUENCE

```
sequence<Query> QueryList;
```

Named by Configuration.

SourcesList SEQUENCE

```
sequence<string> SourcesList;
```

Named by Configuration.

db

3 types.

Fixed records

OutSettings STRUCTURE

Settings for outputs

Name	Type	Description
`name`	string	name of the archive (cannot be empty)
`folder`	string	storage folder
`indexKeyframes`	bool	should we create indexes for keyframes

OutStats STRUCTURE

Statistics of writing

Name	Type	Description
`maxQueueSize`	u64	in bytes

Named by sen.components.recorder.Recorder.

Summary STRUCTURE

Summary of an archive

Name	Type	Description
`firstTime`	sen.TimeStamp	the time of the first event
`lastTime`	sen.TimeStamp	the time of the last event
`keyframeCount`	u32	number of written keyframes
`objectCount`	u32	number of recorded objects
`typeCount`	u32	number of types written
`annotationCount`	u32	number of annotations
`indexedObjectCount`	u32	number of indexed objects

Named by sen.components.replayer.Replay.

kernel

160 types.

Object classes

KernelApi CLASS

Represents the kernel

Name	Type	Flags	Description
`buildInfo`	BuildInfo	RO S BE	kernel build information

Method	Signature	Description
`shutdown`	`()`	requests the kernel to shut down
`getUnits`	`() → UnitList`	registered units
`getTypes`	`() → StringList`	registered types, by name
`getConfig`	`() → KernelParams`	configuration of the kernel execution

VirtualClock CLASS

Allows inspecting and controlling the advance of virtual time and associated component iteration

Name	Type	Flags	Description
`time`	sen.TimeStamp	RO D BE	the current virtual time

VirtualKernelClock CLASS

A clock for all the components loaded by the kernel

Method	Signature	Description
`processNoFlush`	`(delta: sen.Duration) → sen.Duration`	updates the time, drains the inputs, and cycles components - doesn't flush outputs
`flushOutputs`	`() → sen.Duration`	makes component outputs visible to other components and returns the next step duration

VirtualMasterClock CLASS

Controls all the VirtualKernelClock that it discovers in the bus where it's published. One step reaches the next component due to run: each clock reports how long until its own next one, and the soonest wins

Method	Signature	Description
`advanceTime`	`(duration: sen.Duration) → sen.Duration`	steps until the virtual time advanced reaches duration, rounding up to a whole step. Returns the wall time the call took
`step`	`() → sen.Duration`	single step forward
`steps`	`(count: u64) → sen.Duration`	perform 'count' steps forward

Fixed records

AliasTypeSpec STRUCTURE

data of an aliased type

Name	Type	Description
aliasedType	string	qualified type name being aliased

Named by CustomTypeData.

AliasTypeSpecV4 STRUCTURE

Name	Type	Description
aliasedType	string	qualified type name being aliased

Named by CustomTypeDataV4.

AliasTypeSpecV5 STRUCTURE

Name	Type	Description
aliasedType	string	qualified type name being aliased
hash	u32	unique hash

Named by CustomTypeDataV5.

ArgSpec STRUCTURE

an argument of a callable (methods and events)

Name	Type	Description
name	string	argument name
description	string	documentation
type	string	qualified type name of the argument

Named by ArgSpecList.

ArgSpecV4 STRUCTURE

Name	Type	Description
name	string	argument name
description	string	documentation
type	string	qualified type name of the argument

Named by ArgSpecListV4.

ArgSpecV5 STRUCTURE

Name	Type	Description
name	string	argument name

Name	Type	Description
nameHash	u32	hash computed with the argument name
description	string	documentation
type	string	qualified type name of the argument
hash	u32	unique hash

Named by ArgSpecListV5.

BuildInfo STRUCTURE

Build-related information

Name	Type	Description
maintainer	string	principal maintainer of the software
version	string	version string (format-agnostic)
compiler	string	vendor-specific compiler string
debugMode	bool	compiled in debug mode or not
buildTime	string	the commit's timestamp, ISO-8601; not the compiler's clock
wordSize	WordSize	architecture
gitRef	string	git ref spec
gitHash	string	git hash
gitStatus	GitStatus	git status

Named by ComponentInfo, KernelApi.

BuiltComponentParams STRUCTURE

Parameters related to components build by the kernel

Name	Type	Description
name	string	the name of the component
period	sen.Duration	the execution period
config	ComponentConfig	the configuration of the component
imports	StringList	the imported packages

Named by BuiltComponentList.

BusAddress STRUCTURE

Address to connect to a bus

Name	Type	Description
sessionName	string	
busName	string	

ClassTypeSpec STRUCTURE

spec of a class

Name	Type	Description
properties	PropertySpecList	
methods	MethodSpecList	
events	EventSpecList	
constructor	MethodSpec	
parents	StringList	
isInterface	bool	

Named by CustomTypeData.

`ClassTypeSpecV4` STRUCTURE

Name	Type	Description
`properties`	PropertySpecListV4	
`methods`	MethodSpecListV4	
`events`	EventSpecListV4	
`isInterface`	bool	
`id`	u32	
`constructor`	MethodSpecV4	
`parents`	StringList	

Named by CustomTypeDataV4.

`ClassTypeSpecV5` STRUCTURE

Name	Type	Description
`properties`	PropertySpecListV5	
`methods`	MethodSpecListV5	
`events`	EventSpecListV5	
`isInterface`	bool	
`id`	u32	
`constructor`	MethodSpecV5	
`parents`	StringList	
`hash`	u32	

Named by CustomTypeDataV5.

`ComponentConfig` STRUCTURE

Basic runtime configuration for a component

Name	Type	Description
`priority`	Priority	thread priority
`stackSize`	u32	thread stack size in bytes, 0 means default
`group`	u32	the group where to run the component
`cpuAffinity`	u64	CPU affinity mask
`inQueue`	QueueConfig	queuing of inbound information
`outQueue`	QueueConfig	queuing of outbound information
`sleepPolicy`	SleepPolicy	configurable component sleep policy

Named by BuiltComponentParams, LoadedComponentParams.

`ComponentInfo` STRUCTURE

Basic information of a kernel component

Name	Type	Description
`name`	string	
`description`	string	
`buildInfo`	BuildInfo	

`CustomTypeSpec` STRUCTURE

spec of a custom type

Name	Type	Description
name	string	name of the type
qualifiedName	string	package-qualified name
description	string	documentation
data	CustomTypeData	spec data

Named by sen.components.jsonrpc.TypeLookupResult, CustomTypeSpecList.

CustomTypeSpecV4 STRUCTURE

Name	Type	Description
name	string	name of the type
qualifiedName	string	package-qualified name
description	string	documentation
data	CustomTypeDataV4	spec data

Named by CustomTypeSpecListV4.

CustomTypeSpecV5 STRUCTURE

Name	Type	Description
name	string	name of the type
qualifiedName	string	package-qualified name
description	string	documentation
data	CustomTypeDataV5	spec data

Named by CustomTypeSpecListV5.

EnumTypeSpec STRUCTURE

spec of an enum type

Name	Type	Description
enums	EnumeratorSpecList	enumerators
storageType	IntegralType	type used to store the value

Named by CustomTypeData.

EnumTypeSpecV4 STRUCTURE

Name	Type	Description
enums	EnumeratorSpecListV4	enumerators
storageType	IntegralType	type used to store the value

Named by CustomTypeDataV4.

EnumTypeSpecV5 STRUCTURE

Name	Type	Description
enums	EnumeratorSpecListV5	enumerators
storageType	IntegralType	type used to store the value
hash	u32	unique hash

Named by CustomTypeDataV5.

EnumeratorSpec STRUCTURE

an enumerator

Name	Type	Description
name	string	enumerator name
key	u32	enumerator value
description	string	enumerator description

Named by EnumeratorSpecList.

EnumeratorSpecV4 STRUCTURE

Name	Type	Description
name	string	enumerator name
key	u64	enumerator value

Named by EnumeratorSpecListV4.

EnumeratorSpecV5 STRUCTURE

Name	Type	Description
name	string	enumerator name
key	u64	enumerator value
hash	u32	unique hash

Named by EnumeratorSpecListV5.

EnvVar STRUCTURE

Information about an environment variable

Name	Type	Description
name	string	name of the variable
value	string	value of the variable

Named by EnvironmentVarList.

ErrorData STRUCTURE

Information related to the error

Name	Type	Description
time	sen.TimeStamp	time of the error
errorMessage	StringList	message presented to the user
exceptionData	UncaughtException	information about the unhandled exception (if any)
signalData	SignalData	information about the received signal (if any)

Named by ErrorReport.

ErrorReport STRUCTURE

Information about a detected error

Name	Type	Description
senData	SenData	information related to Sen
processData	ProcessData	information related to the process
errorData	ErrorData	information related to the error

EventSpec STRUCTURE

spec for an event

Name	Type	Description
name	string	name of the type
description	string	documentation
args	ArgSpecList	arguments
transportMode	TransportModeSpec	transport mode

Named by EventSpecList.

EventSpecV4 STRUCTURE

Name	Type	Description
memberHash	u32	hash of the class member
name	string	name of the type
description	string	documentation
args	ArgSpecListV4	arguments
transportMode	TransportModeSpec	transport mode

Named by EventSpecListV4.

EventSpecV5 STRUCTURE

Name	Type	Description
memberHash	u32	hash of the class member
name	string	name of the type
description	string	documentation
args	ArgSpecListV5	arguments
transportMode	TransportModeSpec	transport mode
hash	u32	unique hash

Named by EventSpecListV5.

ExecError STRUCTURE

Holds details of an execution error

Name	Type	Description
category	ErrorCategory	general problem
explanation	string	human-readable explanation

KernelParams STRUCTURE

Parameters to configure kernel execution

Name	Type	Description
runMode	RunMode	how to run the kernel
appName	string	optional name of the application
bus	string	where to publish the kernel objects (defaults to local.kernel)
clockBus	string	where to publish the virtual clock (if any), defaults to bus
clockName	string	the name of the virtual clock (if any), defaults to 'clock'
clockMaster	bool	if time is virtualized, publish a master clock to clockBus
logConfig	log.Config	logging configuration
crashReportDir	string	where to store crash reports (defaults to the temp dir)

Name	Type	Description
`crashReportDisabled`	bool	if true, no reports are generated
`lockMemoryPages`	bool	keep process pages memory-resident
`sleepPolicy`	SleepPolicy	configurable sleep policy of the kernel component
`compatibilityMode`	CompatibilityMode	what to do with a remote type that does not match
`limitCpuIdleLatency`	bool	This struct is reachable over the wire through Kernel.getConfig() and is embedded in SenData, and the encoding is positional, so adding a field here shifts what follows it either way. keep the CPUs out of the deep idle states they are slow to leave

Named by KernelApi, SenData.

`LoadedComponentParams` STRUCTURE

Parameters related to a component that will be loaded by the kernel

Name	Type	Description
`name`	string	the name of the component
`config`	ComponentConfig	the configuration of the component

Named by LoadedComponentList.

`MethodSpec` STRUCTURE

spec for methods

Name	Type	Description
`name`	string	name of the type
`description`	string	documentation
`args`	ArgSpecList	arguments
`transportMode`	TransportModeSpec	transport mode
`constness`	MethodConstness Spec	if it changes the state of the object
`deferred`	bool	if the execution of the method is deferred
`returnType`	string	qualified return type name
`propertyRelation`	PropertyRelationSpec	property relation of the method
`localOnly`	bool	true if the method can only be called locally

Named by ClassTypeSpec, MethodSpecList.

`MethodSpecV4` STRUCTURE

Name	Type	Description
`memberHash`	u32	hash of the class member
`name`	string	name of the type
`description`	string	documentation
`args`	ArgSpecListV4	arguments
`transportMode`	TransportModeSpec	transport mode
`constness`	MethodConstness Spec	if it changes the state of the object
`deferred`	bool	true if the method is marked as deferred
`returnTypeId`	string	qualified return type name (empty for none)

Named by ClassTypeSpecV4, MethodSpecListV4.

MethodSpecV5 STRUCTURE

Name	Type	Description
memberHash	u32	hash of the class member
name	string	name of the type
description	string	documentation
args	ArgSpecListV5	arguments
transportMode	TransportModeSpec	transport mode
constness	MethodConstness Spec	if it changes the state of the object
localOnly	bool	true if the method can only be called locally
returnTypeId	string	qualified return type name (empty for none)
hash	u32	unique hash

Named by ClassTypeSpecV5, MethodSpecListV5.

NetworkFootprint STRUCTURE

Network endpoints predicted offline or observed at runtime for one process

Name	Type	Description
discoveryPort	u16	Discovery port used to calculate multicast addresses
multicast	NetworkFootprint Multicast	Multicast allocation details
ports	NetworkFootprintPort List	Port settings
portExclusions	NetworkFootprintPort Exclusions	Port ranges that cannot be used

NetworkFootprintAddressRange STRUCTURE

Inclusive IPv4 address range excluded from multicast allocation

Name	Type	Description
min	string	
max	string	

Named by NetworkFootprintAddressRangeList.

NetworkFootprintBus STRUCTURE

Multicast endpoint allocated to a bus

Name	Type	Description
sessionName	string	
busName	string	
sessionId	u32	
busId	u32	
groupAddress	string	
source	NetworkFootprintBus Source	

Named by NetworkFootprintBusList.

NetworkFootprintBusIdentity STRUCTURE

Names and IDs that identify a bus

Name	Type	Description
sessionName	string	
busName	string	
sessionId	u32	
busId	u32	

Named by NetworkFootprintSelfCollision.

NetworkFootprintCollision STRUCTURE

Multicast allocation capacity and collision information

Name	Type	Description
usableSpaceSize	u64	Number of addresses available
busCount	u64	Number of reported buses
probability	f64	Estimated conflict from 0 to 1
selfCollisions	NetworkFootprintSelfCollisionList	Collisions in the reported buses

Named by NetworkFootprintMulticast.

NetworkFootprintMulticast STRUCTURE

Multicast information in the network footprint

Name	Type	Description
port	u16	Shared multicast port used by every bus
buses	NetworkFootprintBusList	Multicast address used by each bus
exclusions	NetworkFootprintAddressRangeList	Addresses that cannot be used
collision	NetworkFootprintCollision	Available addresses and conflict information

Named by NetworkFootprint.

NetworkFootprintPort STRUCTURE

Socket type, port selection mode and port value

Name	Type	Description
kind	NetworkFootprintPortKind	
mode	NetworkFootprintPortMode	
value	NetworkFootprintPortValue	

Named by NetworkFootprintPortList.

NetworkFootprintPortExclusions STRUCTURE

Port ranges that cannot be used

Name	Type	Description
builtIn	NetworkFootprintPortRangeList	
configured	NetworkFootprintPortRangeList	

Name	Type	Description
`operatingSystem`	NetworkFootprintPort RangeList	

Named by NetworkFootprint.

`NetworkFootprintPortRange` STRUCTURE

Inclusive range from which a port may be selected

Name	Type	Description
`min`	u16	
`max`	u16	

Named by NetworkFootprintPortRangeList, NetworkFootprintPortValue.

`NetworkFootprintSelfCollision` STRUCTURE

Two buses allocated to the same multicast group

Name	Type	Description
`firstBus`	NetworkFootprintBus Identity	
`secondBus`	NetworkFootprintBus Identity	
`groupAddress`	string	

Named by NetworkFootprintSelfCollisionList.

`OpFinished` STRUCTURE

An operation is complete

Named by OpState.

`OpNotFinished` STRUCTURE

An operation is not complete

Name	Type	Description
`waitHint`	sen.Duration	How long to wait until the next try

Named by OpState.

`OptionalTypeSpec` STRUCTURE

data of an optional type

Name	Type	Description
`type`	string	qualified type name

Named by CustomTypeData.

`OptionalTypeSpecV4` STRUCTURE

Name	Type	Description
`type`	string	qualified type name

Named by CustomTypeDataV4.

OptionalTypeSpecV5 STRUCTURE

Name	Type	Description
type	string	qualified type name
hash	u32	unique hash

Named by CustomTypeDataV5.

PrecisionSleep STRUCTURE

Be more precise at the expense of some CPU cycles.

Name	Type	Description
veryCoarseGrainSleepTime	sen.Duration	First set of sleeps. If 0 defaults to 7ms
coarseGrainSleepTime	sen.Duration	Second set of sleeps. If 0 defaults to 1ms

Named by SleepPolicy.

ProcessData STRUCTURE

Information about the process

Name	Type	Description
processInfo	ProcessInfo	general process information
environment	EnvironmentVarList	list of environment variables
stacktrace	StringList	the stack trace (if any)
recentLogs	StringList	the last logs emitted by the process

Named by ErrorReport.

ProcessInfo STRUCTURE

Basic information about a process

Name	Type	Description
hostId	u32	
processId	u32	
sessionId	u32	
sessionName	string	
appName	string	
hostName	string	
osKind	OsKind	
osName	string	
cpuArch	CpuArch	

Named by sen.components.ether.Hello, sen.components.ether.SessionPresenceBeam, ProcessData, Processes.

PropertySpec STRUCTURE

spec of a property

Name	Type	Description
name	string	name of the type
description	string	documentation
category	PropertyCategorySpec	category
type	string	qualified type name of the property
transportMode	TransportModeSpec	how to transport the data

Name	Type	Description
tags	StringList	user-defined tags
checkedSet	bool	if setting the property is checked

Named by PropertySpecList.

PropertySpecV4 STRUCTURE

Name	Type	Description
memberHash	u32	hash of the class member
name	string	name of the type
description	string	documentation
category	PropertyCategorySpec	category
type	string	qualified type name of the property
transportMode	TransportModeSpec	how to transport the data
tags	StringList	user-defined tags

Named by PropertySpecListV4.

PropertySpecV5 STRUCTURE

Name	Type	Description
memberHash	u32	hash of the class member
name	string	name of the type
description	string	documentation
category	PropertyCategorySpec	category
type	string	qualified type name of the property
transportMode	TransportModeSpec	how to transport the data
tags	StringList	user-defined tags
hash	u32	unique hash

Named by PropertySpecListV5.

QuantityTypeSpec STRUCTURE

spec of a quantity type

Name	Type	Description
elementType	NumericType	type used to store the value
unit	UnitInfo	used unit
minValue	f64	discarded if >= maxValue
maxValue	f64	discarded if <= minValue

Named by CustomTypeData.

QuantityTypeSpecV4 STRUCTURE

Name	Type	Description
numericType	NumericType	type used to store the value
unit	UnitInfo	used unit
minValue	f64	discarded if >= maxValue
maxValue	f64	discarded if <= minValue

Named by CustomTypeDataV4.

QuantityTypeSpecV5 STRUCTURE

Name	Type	Description
numericType	NumericType	type used to store the value
unit	UnitInfo	used unit
minValue	f64	discarded if >= maxValue
maxValue	f64	discarded if <= minValue
hash	u32	unique hash

Named by CustomTypeDataV5.

QueueConfig STRUCTURE

Name	Type	Description
evictionPolicy	QueueEvictionPolicy	what to do when the queue is full
maxSize	u64	0 means unbounded

Named by ComponentConfig.

SenData STRUCTURE

Information about Sen

Name	Type	Description
version	string	the Sen version
compiler	string	the compiler used to build Sen
debugMode	bool	true if it was build in debug mode
buildTime	string	the commit's timestamp, not the compiler's clock
wordSize	WordSize	architecture
gitRef	string	git ref
gitHash	string	git hash
gitStatus	GitStatus	status of the branch when build
kernelParams	KernelParams	Sen kernel parameters
kernelProtocol	u32	version of the kernel protocol
transportProtocol	u32	version of the transport protocol (if present)
loadedComponents	LoadedComponentList	components to load
builtComponents	BuiltComponentList	components to build

Named by ErrorReport.

SequenceTypeSpec STRUCTURE

spec of a sequence type

Name	Type	Description
elementType	string	qualified type name for stored values
maxSize	u64	discarded when 0
fixedSize	bool	

Named by CustomTypeData.

SequenceTypeSpecV4 STRUCTURE

Name	Type	Description
elementType	string	qualified type name for stored values

Name	Type	Description
`maxSize`	u64	discarded when 0

Named by CustomTypeDataV4.

`SequenceTypeSpecV5` STRUCTURE

Name	Type	Description
`elementType`	string	qualified type name for stored values
`maxSize`	u64	discarded when 0
`fixedSize`	bool	
`hash`	u32	unique hash

Named by CustomTypeDataV5.

`SignalData` STRUCTURE

Information about an unhandled signal

Name	Type	Description
`signalNumber`	i32	signal number
`signalName`	string	human-readable name

Named by ErrorData.

`StructTypeFieldSpec` STRUCTURE

a field of a struct

Name	Type	Description
`name`	string	field name
`description`	string	documentation
`type`	string	qualified type name held by this field

Named by StructTypeFieldSpecList.

`StructTypeFieldSpecV4` STRUCTURE

Name	Type	Description
`name`	string	field name
`description`	string	documentation
`type`	string	qualified type name held by this field

Named by StructTypeFieldSpecListV4.

`StructTypeFieldSpecV5` STRUCTURE

Name	Type	Description
`name`	string	field name
`description`	string	documentation
`type`	string	qualified type name held by this field
`hash`	u32	unique hash

Named by StructTypeFieldSpecListV5.

StructTypeSpec STRUCTURE

spec of an struct type

Name	Type	Description
fields	StructTypeFieldSpec List	fields
parent	string	qualified type name the parent struct, empty means none

Named by CustomTypeData.

StructTypeSpecV4 STRUCTURE

Name	Type	Description
fields	StructTypeFieldSpec ListV4	fields
parent	string	qualified type name the parent struct, empty means none

Named by CustomTypeDataV4.

StructTypeSpecV5 STRUCTURE

Name	Type	Description
fields	StructTypeFieldSpec ListV5	fields
parent	string	qualified type name the parent struct, empty means none
hash	u32	unique hash

Named by CustomTypeDataV5.

SystemSleep STRUCTURE

Use the native system sleep

Named by SleepPolicy.

UncaughtException STRUCTURE

Information about an unhandled exception

Name	Type	Description
exceptionKind	ExceptionKind	kind of exception
message	string	message it provides (if any)

Named by ErrorData.

UnitInfo STRUCTURE

Describes a unit

Name	Type	Description
name	string	
abbreviation	string	
category	UnitCat	

Named by QuantityTypeSpec, QuantityTypeSpecV4, QuantityTypeSpecV5, UnitList.

VariantTypeFieldSpec STRUCTURE

a field of a variant

Name	Type	Description
key	u32	key/index of this type within the variant
description	string	documentation
type	string	qualified type name held by this field

Named by VariantTypeFieldSpecList.

VariantTypeFieldSpecV4 STRUCTURE

Name	Type	Description
key	u32	key/index of this type within the variant
description	string	documentation
type	string	qualified type name held by this field

Named by VariantTypeFieldSpecListV4.

VariantTypeFieldSpecV5 STRUCTURE

Name	Type	Description
key	u32	key/index of this type within the variant
description	string	documentation
type	string	qualified type name held by this field
hash	u32	unique hash

Named by VariantTypeFieldSpecListV5.

VariantTypeSpec STRUCTURE

spec of an struct type

Name	Type	Description
fields	VariantTypeFieldSpecList	

Named by CustomTypeData.

VariantTypeSpecV4 STRUCTURE

Name	Type	Description
fields	VariantTypeFieldSpecListV4	

Named by CustomTypeDataV4.

VariantTypeSpecV5 STRUCTURE

Name	Type	Description
fields	VariantTypeFieldSpecListV5	
hash	u32	

Named by CustomTypeDataV5.

Enumerations

BasicType ENUMERATION

built-in others

Held as u8.

booleanType	0
stringType	1
durationType	2
timestampType	3

Named by BuiltInType.

CompatibilityMode ENUMERATION

What to do when a type published by another kernel does not match the local one

Held as u8.

relaxed	0
strict	1
disabled	2

Named by KernelParams.

ComponentState ENUMERATION

State of a kernel component

Held as u8.

unloaded	0
preloaded	1
loading	2
loaded	3
initializing	4
initialized	5
running	6
stopping	7
stopped	8
unloading	9
error	10

CpuArch ENUMERATION

Type of CPU

Held as u8.

x86	0
x64	1
arm2	2
arm3	3
arm4T	4
arm5	5
arm6T2	6
arm6	7
arm7	8
arm7a	9
arm7r	10
arm7s	11
arm64	12
mips	13
superH	14
ppc	15
ppc64	16
sparc	17
m68k	18
otherArch	19

Named by ProcessInfo.

`ErrorCategory` ENUMERATION

Category of a problem during component execution

Held as u8.

runtimeError	0
logicError	1
expectationsNotMet	2
ioError	3
other	4

Named by ExecError.

`ExceptionKind` ENUMERATION

Types of exceptions that can occur.

Held as u8.

runtime	0
logic	1
standard	2
unknown	3

Named by UncaughtException.

`GitStatus` ENUMERATION

Status of git

Held as u8.

clean	0
modified	1
unknown	2

Named by BuildInfo, SenData.

`IntegralType` ENUMERATION

built-in integrals

Held as u8.

uint8Type	0	uint32Type	4
int16Type	1	int64Type	5
uint16Type	2	uint64Type	6
int32Type	3		

Named by EnumTypeSpec, EnumTypeSpecV4, EnumTypeSpecV5, NumericType.

MethodConstnessSpec ENUMERATION

method constness

Held as u8.

constant	0
nonConstant	1

Named by MethodSpec, MethodSpecV4, MethodSpecV5.

NetworkFootprintBusSource ENUMERATION

Bus source

Held as u8.

configured	0
supplied	1
runtime	2

Named by NetworkFootprintBus.

NetworkFootprintPortKind ENUMERATION

Type of socket opened by the process

Held as u8.

tcpAcceptor	0
udpUnicast	1
tcpSource	2

Named by NetworkFootprintPort.

NetworkFootprintPortMode ENUMERATION

Port allocation policy

Held as u8.

ephemeral	0
probe	1
pinned	2

Named by NetworkFootprintPort.

OsKind ENUMERATION

Type of operating system

Held as u8.

windowsOs	0
linuxOs	1
androidOs	2
appleOs	3
unixOs	4
posixOs	5
otherOs	6

Named by ProcessInfo.

Priority ENUMERATION

Thread execution priorities

Held as u32.

lowest	0	nominalMax	2
nominalMin	1	highest	3

Named by ComponentConfig.

PropertyCategorySpec ENUMERATION

how a property is seen by others

Held as u8.

staticRW	0
staticRO	1
dynamicRW	2
dynamicRO	3

Named by PropertySpec, PropertySpecV4, PropertySpecV5.

PropertyRelationSpec ENUMERATION

method property relation

Held as u8.

nonPropertyRelated	0
propertyGetter	1
propertySetter	2

Named by MethodSpec.

QueueEvictionPolicy ENUMERATION

Component queue eviction policy

Held as u32.

dropOldest	0
dropNewest	1

Named by QueueConfig.

RealType ENUMERATION

built-in reals

Held as u8.

float32Type	0
float64Type	1

Named by NumericType.

RunMode ENUMERATION

Run modes for the kernel

Held as u8.

realTime	0
virtualTime	1
virtualTimeRunning	2
startAndStop	3

Named by KernelParams.

ThreadCreateErr ENUMERATION

What can go wrong when creating a thread

Held as u32.

schedulerAlreadyStarted	0
invalidStackSize	1
invalidThreadFunction	2
invalidThreadFunctionArgument	3
invalidAffinity	4
tooManyThreads	5
internalOsError	6

TransportModeSpec ENUMERATION

how to transport information

Held as u8.

unicast	0
multicast	1
confirmed	2

Named by EventSpec, EventSpecV4, EventSpecV5, MethodSpec, MethodSpecV4, MethodSpecV5, PropertySpec, PropertySpecV4, PropertySpecV5.

UnitCat ENUMERATION

Type of measurement

Held as u8.

length	0
mass	1
time	2
angle	3
temperature	4
frequency	5
velocity	6
angularVelocity	7
acceleration	8
angularAcceleration	9
density	10
pressure	11
area	12
force	13
torque	14

Named by UnitInfo.

WordSize ENUMERATION

Native processor word size

Held as u32.

bits32	0
bits64	1

Named by BuildInfo, SenData.

Variant records

BuiltInType VARIANT

all built-in types

Type	Description
NumericType	
BasicType	

CustomTypeData VARIANT

all custom types

Type	Description
EnumTypeSpec	
QuantityTypeSpec	
SequenceTypeSpec	
StructTypeSpec	
VariantTypeSpec	
AliasTypeSpec	
OptionalTypeSpec	
ClassTypeSpec	

Named by CustomTypeSpec.

CustomTypeDataV4 VARIANT

Type	Description
EnumTypeSpecV4	
QuantityTypeSpecV4	
SequenceTypeSpecV4	
StructTypeSpecV4	
VariantTypeSpecV4	
AliasTypeSpecV4	
OptionalTypeSpecV4	
ClassTypeSpecV4	

Named by CustomTypeSpecV4.

CustomTypeDataV5 VARIANT

Type	Description
EnumTypeSpecV5	
QuantityTypeSpecV5	
SequenceTypeSpecV5	
StructTypeSpecV5	
VariantTypeSpecV5	
AliasTypeSpecV5	
OptionalTypeSpecV5	
ClassTypeSpecV5	

Named by CustomTypeSpecV5.

NetworkFootprintPortValue VARIANT

Known port number or range

Type	Description
u16	
NetworkFootprintPort Range	

Named by NetworkFootprintPort.

NumericType VARIANT

all built-in numeric types

Type	Description
IntegralType	
RealType	

Named by BuiltInType, QuantityTypeSpec, QuantityTypeSpecV4, QuantityTypeSpecV5.

OpState VARIANT

The status of an operation

Type	Description
OpFinished	done
OpNotFinished	not done

SleepPolicy VARIANT

Component sleep policy

Type	Description
PrecisionSleep	
SystemSleep	

Named by ComponentConfig, KernelParams.

Arrays

ArgSpecList SEQUENCE

list of argument specs

```
sequence<ArgSpec> ArgSpecList;
```

Named by EventSpec, MethodSpec.

ArgSpecListV4 SEQUENCE

```
sequence<ArgSpecV4> ArgSpecListV4;
```

Named by EventSpecV4, MethodSpecV4.

ArgSpecListV5 SEQUENCE

```
sequence<ArgSpecV5> ArgSpecListV5;
```

Named by EventSpecV5, MethodSpecV5.

Buffer SEQUENCE

A general-purpose unbounded buffer

```
sequence<u8> Buffer;
```

Named by sen.components.jsonrpc.StaticFile.

BuiltComponentList SEQUENCE

```
sequence<BuiltComponentParams> BuiltComponentList;
```

Named by SenData.

CustomTypeSpecList SEQUENCE

list of custom types

```
sequence<CustomTypeSpec> CustomTypeSpecList;
```

Named by sen.components.jsonrpc.JsonRpcServer.

CustomTypeSpecListV4 SEQUENCE

```
sequence<CustomTypeSpecV4> CustomTypeSpecListV4;
```

CustomTypeSpecListV5 SEQUENCE

```
sequence<CustomTypeSpecV5> CustomTypeSpecListV5;
```

EnumeratorSpecList SEQUENCE

list of enumerators

```
sequence<EnumeratorSpec> EnumeratorSpecList;
```

Named by EnumTypeSpec.

EnumeratorSpecListV4 SEQUENCE

```
sequence<EnumeratorSpecV4> EnumeratorSpecListV4;
```

Named by EnumTypeSpecV4.

EnumeratorSpecListV5 SEQUENCE

```
sequence<EnumeratorSpecV5> EnumeratorSpecListV5;
```

Named by EnumTypeSpecV5.

EnvironmentVarList SEQUENCE

```
sequence<EnvVar> EnvironmentVarList;
```

Named by ProcessData.

EventSpecList SEQUENCE

list of event specs

```
sequence<EventSpec> EventSpecList;
```

Named by ClassTypeSpec.

EventSpecListV4 SEQUENCE

```
sequence<EventSpecV4> EventSpecListV4;
```

Named by ClassTypeSpecV4.

EventSpecListV5 SEQUENCE

```
sequence<EventSpecV5> EventSpecListV5;
```

Named by ClassTypeSpecV5.

LoadedComponentList SEQUENCE

```
sequence<LoadedComponentParams> LoadedComponentList;
```

Named by SenData.

MethodSpecList SEQUENCE

list of method specs

```
sequence<MethodSpec> MethodSpecList;
```

Named by ClassTypeSpec.

MethodSpecListV4 SEQUENCE

```
sequence<MethodSpecV4> MethodSpecListV4;
```

Named by ClassTypeSpecV4.

MethodSpecListV5 `SEQUENCE`

```
sequence<MethodSpecV5> MethodSpecListV5;
```

Named by ClassTypeSpecV5.

NetworkFootprintAddressRangeList `SEQUENCE`

```
sequence<NetworkFootprintAddressRange> NetworkFootprintAddressRangeList;
```

Named by NetworkFootprintMulticast.

NetworkFootprintBusList `SEQUENCE`

Buses included in a network footprint

```
sequence<NetworkFootprintBus> NetworkFootprintBusList;
```

Named by NetworkFootprintMulticast.

NetworkFootprintPortList `SEQUENCE`

Ports used by the process

```
sequence<NetworkFootprintPort> NetworkFootprintPortList;
```

Named by NetworkFootprint.

NetworkFootprintPortRangeList `SEQUENCE`

```
sequence<NetworkFootprintPortRange> NetworkFootprintPortRangeList;
```

Named by NetworkFootprintPortExclusions.

NetworkFootprintSelfCollisionList `SEQUENCE`

Multicast address conflicts found in the reported buses

```
sequence<NetworkFootprintSelfCollision> NetworkFootprintSelfCollisionList;
```

Named by NetworkFootprintCollision.

Processes `SEQUENCE`

Unbounded sequence of ProcessInfo

```
sequence<ProcessInfo> Processes;
```

PropertySpecList `SEQUENCE`

list of property specs

```
sequence<PropertySpec> PropertySpecList;
```

Named by ClassTypeSpec.

PropertySpecListV4 `SEQUENCE`

```
sequence<PropertySpecV4> PropertySpecListV4;
```

Named by ClassTypeSpecV4.

PropertySpecListV5 SEQUENCE

```
sequence<PropertySpecV5> PropertySpecListV5;
```

Named by ClassTypeSpecV5.

StringList SEQUENCE

Unbounded sequence of strings for general purpose

```
sequence<string> StringList;
```

Named by sen.components.jsonrpc.JsonRpcServer, sen.components.jsonrpc.NamedSelection, sen.components.jsonrpc.SessionInfo, BuiltComponentParams, ClassTypeSpec, ClassTypeSpecV4, ClassTypeSpecV5, ErrorData, KernelApi, ProcessData, PropertySpec, PropertySpecV4, PropertySpecV5.

StructTypeFieldSpecList SEQUENCE

list of struct fields

```
sequence<StructTypeFieldSpec> StructTypeFieldSpecList;
```

Named by StructTypeSpec.

StructTypeFieldSpecListV4 SEQUENCE

```
sequence<StructTypeFieldSpecV4> StructTypeFieldSpecListV4;
```

Named by StructTypeSpecV4.

StructTypeFieldSpecListV5 SEQUENCE

```
sequence<StructTypeFieldSpecV5> StructTypeFieldSpecListV5;
```

Named by StructTypeSpecV5.

U32List SEQUENCE

```
sequence<u32> U32List;
```

UnitList SEQUENCE

Unbounded sequence of unit information structs

```
sequence<UnitInfo> UnitList;
```

Named by KernelApi.

VariantTypeFieldSpecList SEQUENCE

list of variant fields

```
sequence<VariantTypeFieldSpec> VariantTypeFieldSpecList;
```

Named by VariantTypeSpec.

VariantTypeFieldSpecListV4 SEQUENCE

```
sequence<VariantTypeFieldSpecV4> VariantTypeFieldSpecListV4;
```

Named by VariantTypeSpecV4.

VariantTypeFieldSpecListV5 SEQUENCE

```
sequence<VariantTypeFieldSpecV5> VariantTypeFieldSpecListV5;
```

Named by VariantTypeSpecV5.

Optionals

MaybeException OPTIONAL

```
optional<UncaughtException> MaybeException;
```

MaybeF64 OPTIONAL

optional value for a f64

```
optional<f64> MaybeF64;
```

MaybeNetworkFootprintMulticast OPTIONAL

```
optional<NetworkFootprintMulticast> MaybeNetworkFootprintMulticast;
```

MaybeNetworkFootprintPortValue OPTIONAL

The value is not present when the port number is not known

```
optional<NetworkFootprintPortValue> MaybeNetworkFootprintPortValue;
```

MaybeSignal OPTIONAL

```
optional<SignalData> MaybeSignal;
```

MaybeStacktrace OPTIONAL

```
optional<StringList> MaybeStacktrace;
```

MaybeTransportProtocol OPTIONAL

```
optional<u32> MaybeTransportProtocol;
```

MaybeU64 OPTIONAL

optional value for a u64

```
optional<u64> MaybeU64;
```

log

16 types.

Fixed records

BasicFile STRUCTURE

Basic file sink

Name	Type	Description
`fileName`	string	file to log to
`truncate`	bool	defaults to false
`createParentDir`	bool	true if we should create the parent dir first

Named by SinkType.

ColorStderr STRUCTURE

Colored stderr sink

Named by SinkType.

ColorStdout STRUCTURE

Colored stdout sink

Named by SinkType.

Config STRUCTURE

Overall component configuration

Name	Type	Description
`pattern`	string	default format (empty means default)
`level`	LogLevel	default global level for all
`sinks`	SinkConfigList	sinks to configure
`loggers`	LoggerConfigList	loggers to configure
`backtrace`	bool	log backtrace is printed on failure

Named by sen.kernel.KernelParams.

LoggerConfig STRUCTURE

Configuration used for a given spdlog logger.

Name	Type	Description
`name`	string	name of the logger
`sinks`	SinkNameList	name of the sinks to connect the logger to
`level`	LogLevel	default log level for this logger

Name	Type	Description
`pattern`	string	Formatting for the sinks in this logger. Does nothing if empty. Note that this replaces the format of the associated sinks, and this could affect other loggers using those sinks. The config is applied following the order of the Config.loggers field.

Named by LoggerConfigList.

`Null` STRUCTURE

Single-threaded null sink

Named by SinkType.

`RotatingFile` STRUCTURE

Rotating file sink

Name	Type	Description
`baseFileName`	string	
`maxSize`	u32	0 means use defaults
`maxFiles`	u32	0 means use defaults

Named by SinkType.

`SinkConfig` STRUCTURE

Configuration used for a given spdlog Sink.

Name	Type	Description
`name`	string	name of the sink
`singleThreaded`	bool	if true, single-threaded (defaults to false)
`pattern`	string	the pattern used for the formatting (empty means default).
`level`	LogLevel	default log level for this sink
`config`	SinkType	the type-specific configuration parameters

Named by SinkConfigList.

`Stderr` STRUCTURE

stderr sink

Named by SinkType.

`Stdout` STRUCTURE

stdout sink

Named by SinkType.

`Syslog` STRUCTURE

Syslog sink (only in linux)

Name	Type	Description
`identity`	string	
`option`	u32	
`enableFormatting`	bool	

Named by SinkType.

Enumerations

LogLevel ENUMERATION

Severity/importance of a log message.

Held as u8.

warn	0
err	1
critical	2
trace	3
debug	4
info	5
off	6

Named by sen.components.logmaster.LogMaster, sen.components.logmaster.Logger, Config, LoggerConfig, SinkConfig.

Variant records

SinkType VARIANT

The type-specific configuration of the used spdlog Sink.

Type	Description
ColorStdout	
ColorStderr	
Stdout	
Stderr	
BasicFile	
RotatingFile	
Null	
Syslog	

Named by SinkConfig.

Arrays

LoggerConfigList SEQUENCE

```
sequence<LoggerConfig> LoggerConfigList;
```

Named by Config.

SinkConfigList SEQUENCE

```
sequence<SinkConfig> SinkConfigList;
```

Named by Config.

SinkNameList SEQUENCE

```
sequence<string> SinkNameList;
```

Named by LoggerConfig.

Built-in types

The types the language provides. Every one is written the same way on the wire wherever it appears.

Type	Meaning
`bool`	classic boolean
`f32`	32 bit floating point number
`f64`	64 bit floating point number
`i32`	32 bit signed integer
`sen.Duration`	a time duration
`sen.TimeStamp`	a point in time
`string`	unbounded string
`u16`	16 bit unsigned integer
`u32`	32 bit unsigned integer
`u64`	64 bit unsigned integer
`u8`	8 bit unsigned integer

Index